

HE2003 Econometrics I: Linear Regression with Multiple Regressors 2

Wenjie Wang

**Economics, School of Social Sciences
Nanyang Technological University**

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Linear Regression with Multiple Regressors

KEY CONCEPT: Multicollinearity

- Multicollinearity arises when **one of the regressors is a perfect linear combination of the other regressors.**
- A famous example of multicollinearity is the so-called **dummy variable trap.**

Let us first consider the following **model with a binary regressor**:

$$wage_i = \beta_0 + \beta_1 D_i + u_i, i = 1, \dots, n$$

where $wage_i$ is the **wage** of i -th person and D_i is a **dummy variable** (binary regressor) that satisfies

$$D_i = \begin{cases} 1 & \text{if } i\text{-th person is female;} \\ 0 & \text{if } i\text{-th person is male.} \end{cases}$$

Then, β_1 is the coefficient for **testing if there is gender gap in wage** (That is, we want to test $H_0 : \beta_1 = 0$).

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In particular, we have learnt that $E[u_i|D_i] = 0$ by OLS Assumption 1 and

$$E[\text{wage}_i|D_i = 1] = \beta_0 + \beta_1,$$

$$E[\text{wage}_i|D_i = 0] = \beta_0.$$

$E[\text{wage}_i|D_i = 1]$ is the average wage of the **group of women**, and $E[\text{wage}_i|D_i = 0]$ the average wage of the **group of men**. Thus, β_1 is the difference in the average wages between women and men:

$$\beta_1 = E[\text{wage}_i|D_i = 1] - E[\text{wage}_i|D_i = 0].$$

Hypothesis testing and Confidence Interval with Dummy Variable

- $H_0 : \beta_1 = 0$ (no gender gap in wage) can be rejected at the 5% level when the t -statistic exceeds 1.96 in absolute value.
- Similarly, a 95% confidence interval for β_1 can be constructed as

$$[\hat{\beta}_1 - 1.96 \times SE(\hat{\beta}_1), \hat{\beta}_1 + 1.96 \times SE(\hat{\beta}_1)].$$

- In sum, the usual procedures remain valid for the model with dummy variable.

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Now, suppose that we modify the model by including **an additional dummy variable**. The new model is

$$wage_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + u_i, i = 1, \dots, n$$

where

$$D_{1i} = \begin{cases} 1 & \text{if } i \text{ is male} \\ 0 & \text{if } i \text{ is female} \end{cases} \quad \text{and} \quad D_{2i} = \begin{cases} 1 & \text{if } i \text{ is female} \\ 0 & \text{if } i \text{ is male.} \end{cases}$$

This model is **not estimable** because of **multicollinearity**, and this is called the **dummy variable trap**. Notice that for all i from 1 to n ,

$$1 = D_{1i} + D_{2i}.$$

In addition, **there is an intercept** in the regression, and the **constant term**, the regressor for intercept, is **equal to one for all i** . Thus, **the regression will fail** because of multicollinearity [One of the regressors is a perfect linear combination of the other regressors (D_{1i}, D_{2i})].

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Examples of Multicollinearity

Example 1: Fraction of English learners. Suppose that $EL_Pct_i = 100 \times EL_Frac_i$ for each i (That is, $X_{2i} = aX_{1i}$), and EL_Frac is included in the regression in addition to STR and EL_Pct , the regressors would be multicollinear (What is the effect of a unit **change in EL_Pct , holding EL_Frac constant?** $\rightarrow$ **Not meaningful**).

Example 2: “Not very small” classes. Let NVS_i be a dummy variable that equals 1 if $STR_i \geq 12$ and equals 0 otherwise. This regression also exhibits multicollinearity in the data. There are in fact no districts in the data set with $STR_i < 12$ (The smallest is 14). Thus, $NVS_i = 1$ for all observations. Recall that there is **an intercept already**...

Question: For the STR and Test Score example, suppose we want to allow for **different intercepts for Urban, Suburb, and Rural areas**, **how many dummy variables should we use?**

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Hypothesis Testing and Confidence Interval for a Single Coefficient (in Multiple Regression)

The model is:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki} + u_i,$$

where $i = 1, \dots, n$.

- Suppose that you want to test the hypothesis that the true coefficient β_j equals some specific value, $\beta_{j,0}$.
- **The procedure is very similar to the single regressor case.**
- The null hypothesis and the two-sided alternative hypothesis are

$$H_0 : \beta_j = \beta_{j,0} \quad \text{vs.} \quad H_1 : \beta_j \neq \beta_{j,0}.$$

where $j = 1, 2, \dots, k$.

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KEY CONCEPT: t -test, p -value, and Confidence Interval for a Single Coefficient (in Multiple Regression)

Under $H_0 : \beta_j = \beta_{j,0}$ and large samples, the t -statistic

$$t_j = \frac{\hat{\beta}_j - \beta_{j,0}}{SE(\hat{\beta}_j)}$$

follows a standard normal distribution $N(0, 1)$, where $SE(\hat{\beta}_j)$ is the standard error of $\hat{\beta}_j$, $j = 1, 2, \dots, k$.

- Reject H_0 if $|t_j^{act}| > 1.96$ for test with 5% significance level,
- Reject H_0 if $p_j\text{-value} < 5\%$ for test with 5% significance level,
- $CI_{\beta_j, 95\%} = \left[\hat{\beta}_j - 1.96 \times SE(\hat{\beta}_j), \hat{\beta}_j + 1.96 \times SE(\hat{\beta}_j) \right]$.

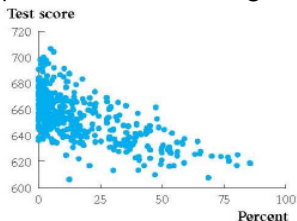
Linear Regression with Multiple Regressors

Application of Multiple Regression to Test Scores

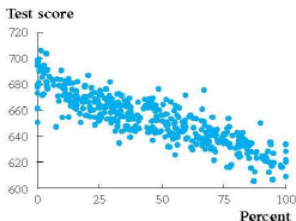
- To solve the OVB problem, we consider three variables (**control variables**) that control for background characteristics of the students that could affect the test scores:
 - (1) **Percentage of English learners (language background, EL_Pct)**,
 - (2) **Percentage of students who are eligible for a reduced price or free lunch in school (economic background, $Meal_Pct$)**,
and
 - (3) **Percentage of students in the district whose families qualify for income assistance program (economic background, $Calw_Pct$)**.
- Can we reject the null hypothesis that a change in STR has no effect on test score ($H_0 : \beta_1 = 0$) **after controlling for these factors?**

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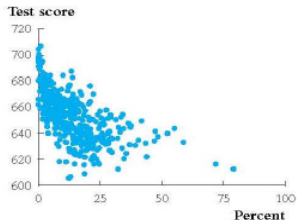
The **scatterplot** shows a **negative relationship** between test scores and percentage of English learners (STATA: `scatter testscr el_pct`). Similar pattern for other background variables.



(a) Percentage of English language learners



(b) Percentage qualifying for reduced price lunch



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Controlling for the percentage of English learners:

$t^{act} = -1.1/0.43 = -2.54$ and p -value is equal to 1.1%.

```
. reg testscr str el_pct, robust
```

Linear regression

Number of obs = 420

F(2, 417) = 223.82

Prob > F = 0.0000

R-squared = 0.4264

Root MSE = 14.464

testscr	Robust					
	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
str	-1.101296	.4328472	-2.54	0.011	-1.95213	-.2504616
el_pct	-.6497768	.0310318	-20.94	0.000	-.710775	-.5887786
_cons	686.0322	8.728224	78.60	0.000	668.8754	703.189

Linear Regression with Multiple Regressors

Controlling for the percentage of English learners and percentage of students eligible for subsidized lunch:

```
. reg testscr str el_pct meal_pct, robust
```

Linear regression

Number of obs = 420
F(3, 416) = 453.48
Prob > F = 0.0000
R-squared = 0.7745
Root MSE = 9.0801

testscr	Robust					
	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
str	-.9983092	.2700799	-3.70	0.000	-1.529201	-.4674178
el_pct	-.1215733	.0328317	-3.70	0.000	-.18611	-.0570366
meal_pct	-.5473456	.0241072	-22.70	0.000	-.5947328	-.4999583
_cons	700.15	5.56845	125.74	0.000	689.2042	711.0958

Linear Regression with Multiple Regressors

Controlling for the percentage of English learners and percentage of students on public income assistance:

```
. reg testscr str el_pct calw_pct, robust
```

Linear regression

Number of obs = 420

F(3, 416) = 170.37

Prob > F = 0.0000

R-squared = 0.6285

Root MSE = 11.654

testscr	Robust					[95% Conf. Interval]
	Coef.	Std. Err.	t	P> t		
str	-1.307984	.3390761	-3.86	0.000	-1.9745	-.641468
el_pct	-.4876202	.0295823	-16.48	0.000	-.5457696	-.4294708
calw_pct	-.7899648	.0676596	-11.68	0.000	-.9229622	-.6569674
_cons	697.9987	6.920369	100.86	0.000	684.3954	711.6019

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Summary of Findings:

- **Controlling for these variables** (language and economic background) **cuts the effect of STR into half** [Therefore, there is probably **downward OVB** in the simple regression with single regressor STR].
- **Robustness check**: The estimation result for the coefficient on STR is not sensitive to which specific control variables are included ($\hat{\beta}_1$ always around -1). In all cases the coefficient remains **statistically significant at at least 5% level**.
- The student background variables are **important predictors of test scores** (They **explain a large fraction of the variation in test scores**): Remember that the R^2 with the single regressor STR is only around 0.05. The $R^2 (= ESS/TSS)$ **jumps** when the background variables are added. [However, note that **R^2 and t -test/ p -value answer very different questions**: Here, t -test/ p -value answers if (after controlling for other factors) $\beta_1 = \Delta \text{TestScore} / \Delta \text{STR}$ is statistically different from zero (which is exactly the superintendent wants to know for her policy), while R^2 cannot answer this question.]